Avi Ignaczak

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EDUCATION

Purdue University, Mitch Daniels School of Business — **West Lafayette, IN**
Integrated Business and Engineering, Mechanical Engineering Concentration — **AUG 2023 - Present**

- VIP Learning Community

PROFESSIONAL EXPERIENCE

Phillips Corporation, Federal Division – Columbia Class Submarines — **Danville, VA | MAY 2025 – AUG 2025, MAY 2026 – AUG 2026**
Additive Manufacturing Intern

- Assisted with the manufacturing process to create parts for the **Columbia Class Submarines** for the **US Navy**.
- Simulated and assessed **thermal warping** conditions for DED machines and created a report on solutions.
- Learned and operated **PBF** and **DED** additive manufacturing processes.
- Researched the effects of **transient conditions** verses **static loads** as it pertains to **heat transfer**.

Bechtel Innovation Design Center – Purdue University — **West Lafayette, IN**
Teaching Assistant — **AUG 2025 - Present**

- Approved and validated student **CAD** and **CAM** files for manufacturability and compliance with lab standards.
- Operated and maintained **CNC mills and lathes**, ensuring safe operation, precision, and consistent machining quality.
- Mentored students through the **design-to-manufacturing process**, providing hands-on support with CNC programming, tooling, setup, and troubleshooting.
- Trained students on **machining best practices, shop safety, and proper equipment operation** to improve manufacturing outcomes.

American Society of Mechanical Engineers Biomechanics Club — **West Lafayette, IN**
Chief Engineer — **JAN 2025 - Present**

- Designed and prototyped a **lightweight** and **portable traction splint** using CAD software and 3D Printing.
- Collaborated with team members to ensure effective communication throughout the **design process**.
- Utilized **Generative Design** to iterate through models and create a lightweight part to satisfy the design criteria.
- Taught new members of the group how to use **Fusion 360** correctly and efficiently.

Avi's 3D Printing — **Canton, MI**
Owner — **FEB 2020 – DEC 2023**

- Founded a 3D printing business during the COVID-19 pandemic, manufacturing mask strain-relief devices for healthcare workers.
- Expanded production capacity from **1 to 5 3D printers** to meet growing customer demand.
- Scaled production from **10 to 120 units per month**, generating up to **$5,000 in monthly sales**.
- Managed the full product lifecycle, including **design, manufacturing, quality control, order fulfillment, and customer communication**.

COMMUNITY SERVICE/VOLUNTEERISM

- Created a virtual tool library in Fusion 360 for 180 students at my High School during my time off from school and it set up a foundation for students to learn about CNC Milling and machining.

RELEVANT SKILLS/CERTIFICATIONS

- Autodesk Fusion 360, Additive Manufacturing, CNC Mill/ Lathe, MIG Welding, MATLAB, Python, OSHA 10 Certification, Finite Element Analysis, Generative Design, Composite Manufacturing, Electrical Systems.

PROJECTS

- **Crosskart – Independent Suspension Race Car**
  Initiated and led a newly established independent study at my high school focused on building an independent suspension race car. Gained hands-on experience in project management, team leadership, and mechanical design.

LEADERSHIP AND INVOLVEMENT

- American Society of Mechanical Engineers, Theta Chi Racing Team (Purdue Grand Prix), The Network for Engineering and Business, Theta Chi Fraternity, IBE EV Racing Team, SAE Baja Team
  Currently refining and redesigning key components to improve performance and reliability.